TAM Airlines Flight 3054

Flight model: Airbus A320
Date: July 17, 2007

From: Porto Alegre, Brazil
To: São Paulo, Brazil

São Paulo airport is situated in the middle of the city surrounded by apartments and roads. Runway 35L is built on a hill top making it very challenging. Also at São Paulo, the weather condition was rainy.

Problem

The pilots were informed that one of the 2 thrust reverses were not working. The runway is slippery and wet. The pilot takes control of the plane and turns off the auto pilot. The pilot lands the plane and applies the reverse thrust but, the plane fails to slow down. Then the pilots tries to bring the plane to stop condition but the plane turns to the left and falls down the hill ans crashes.

Investigation

CCTV camera footage show that the plane was traveling 3 times faster than usual as the time of landing. The runway was wet a the time of landing which would create a layer between the wheel and surface of the water called as **hydroplaning**, which make pilot not able to control the plane. The runway 35L was reported to be very slippery and had been resurfaced just few months back. But, the resurfacing did not include grooving which was scheduled for a later date. These **grooves** carry away rain water from the runway. The thrust reverses were deactivated for maintenance 4 days back and had landed safely in many airports including in São Paulo airport runway 35L earlier. The DVR and FDR contains a temperature chip which turns color when exposed to higher threshold of heat. The FDR and DVR managed to survive the heat, the data from the FDR shows that instead the left engine were in reverse but the right engine were powering up to take off conditions which is opposite to what is expected for a landing. With one engine full power the plane was unable to stop making the plane to turn to the left. An **AFU (Artificial feel unit)** is used to control the thrust using the thrust levers, investigators look for any fault in the AFU. But, find no evidence and rules out any mechanical failure. The crew finally concludes that the pilots would have brought the left engine to reversal but, right engine were in full power and were not brought to reversal or idle position. There is an old procedure to land with a single engine where in both engines are brought to idle and the working reversal engine is brought to reversal, in case of the new procedure both the engines must be brought to idle followed by reversal. But, here pilots failed to bring the right engine to idle position. It is also found that the old procedure was prone to pilot error but, the older procedure could bring the plane to stop much earlier than newer procedure which might be the reason behind the pilots using it to land in runway 35L. Pilots had lots of obstacles along with the difficulty of runway 35L. A real time simulation showed that the position of the thrust levers cannot be brought to notice in a low lit cockpit.

Changes that were brought after the incident

Airbus had changed that both engines would come down to idle procedure and after touch down both would come to reversal procedure. Also, the runway 34L had been grooved. Many pilots still find difficult to land in runway 35L in rainy conditions.

Reference

1. The Airbus A320 That Couldn't Stop — Mayday: Air Disaster
   `https://www.youtube.com/watch?v=0ZrUIzLh27A`